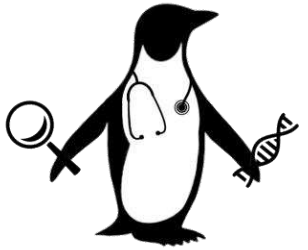
GHENT UNIVERSITY

IDLab
INTERNET & DATA LAB

vito

fwo



PENGQUIN

Personal Genomic Data Sharing:

Current Challenges & Future Possibilities

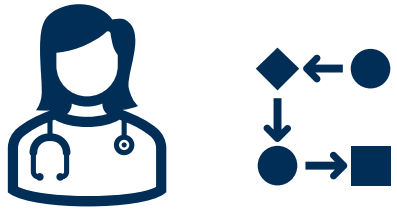
Elias Crum

VITO Digital Precision Health Day
October 10th, 2025

© Elias Crum 2025

The Main Challenges (for clinical use)

Clinical Workflow Integration



The Genomic Data



The Regulatory Landscape



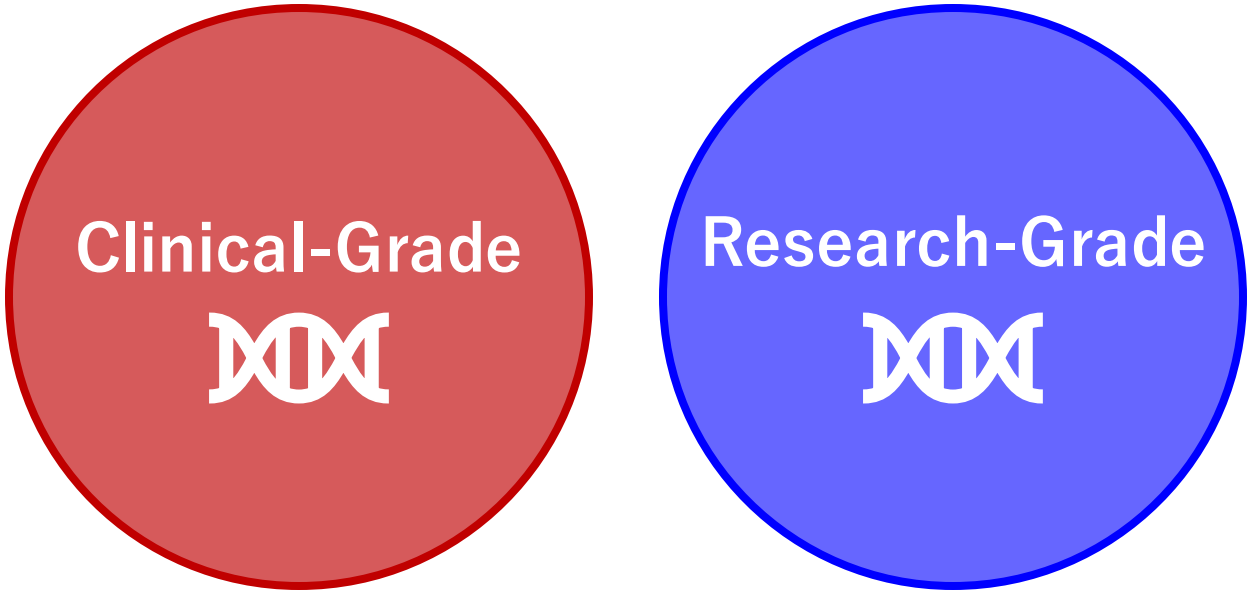
Technical Implementation



© Elias Crum 2025

Genomic Data for Clinical Use

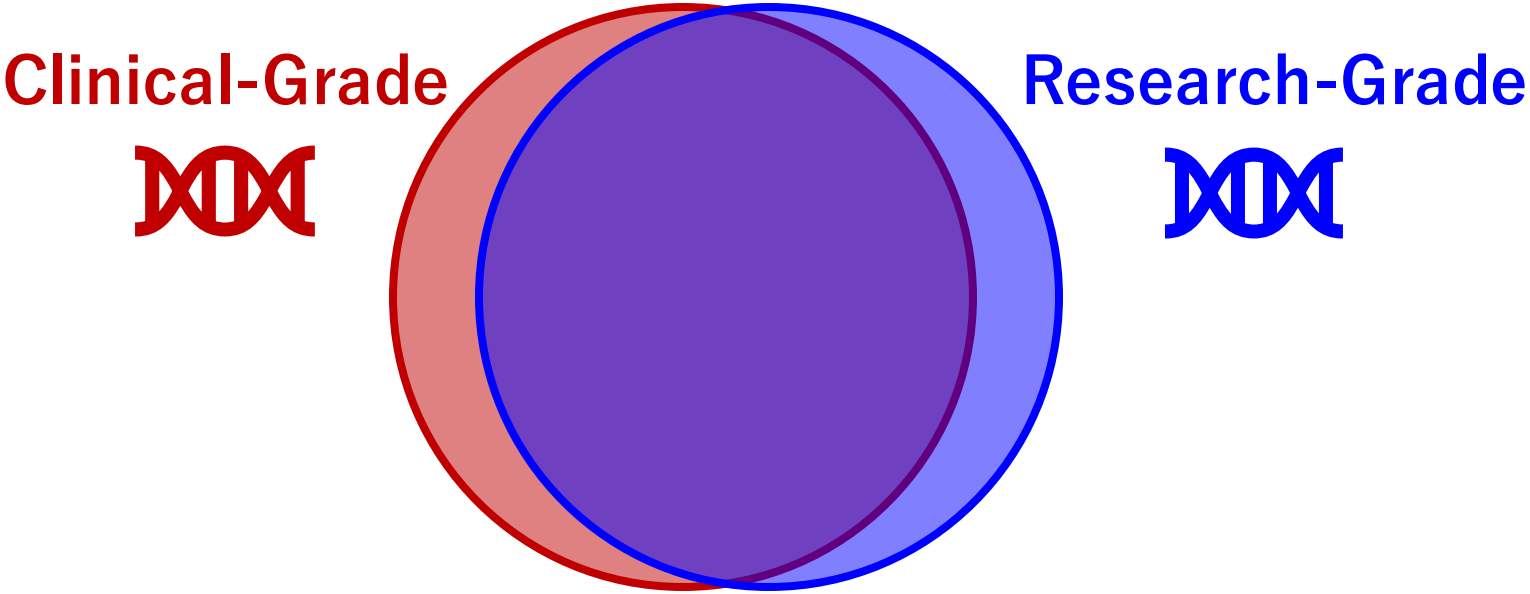
- Format is use-case specific
- Whole Genome Sequence (WGS) data
- Large Projects: 1+ Million Genomes (EU) [1]



[1] https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes

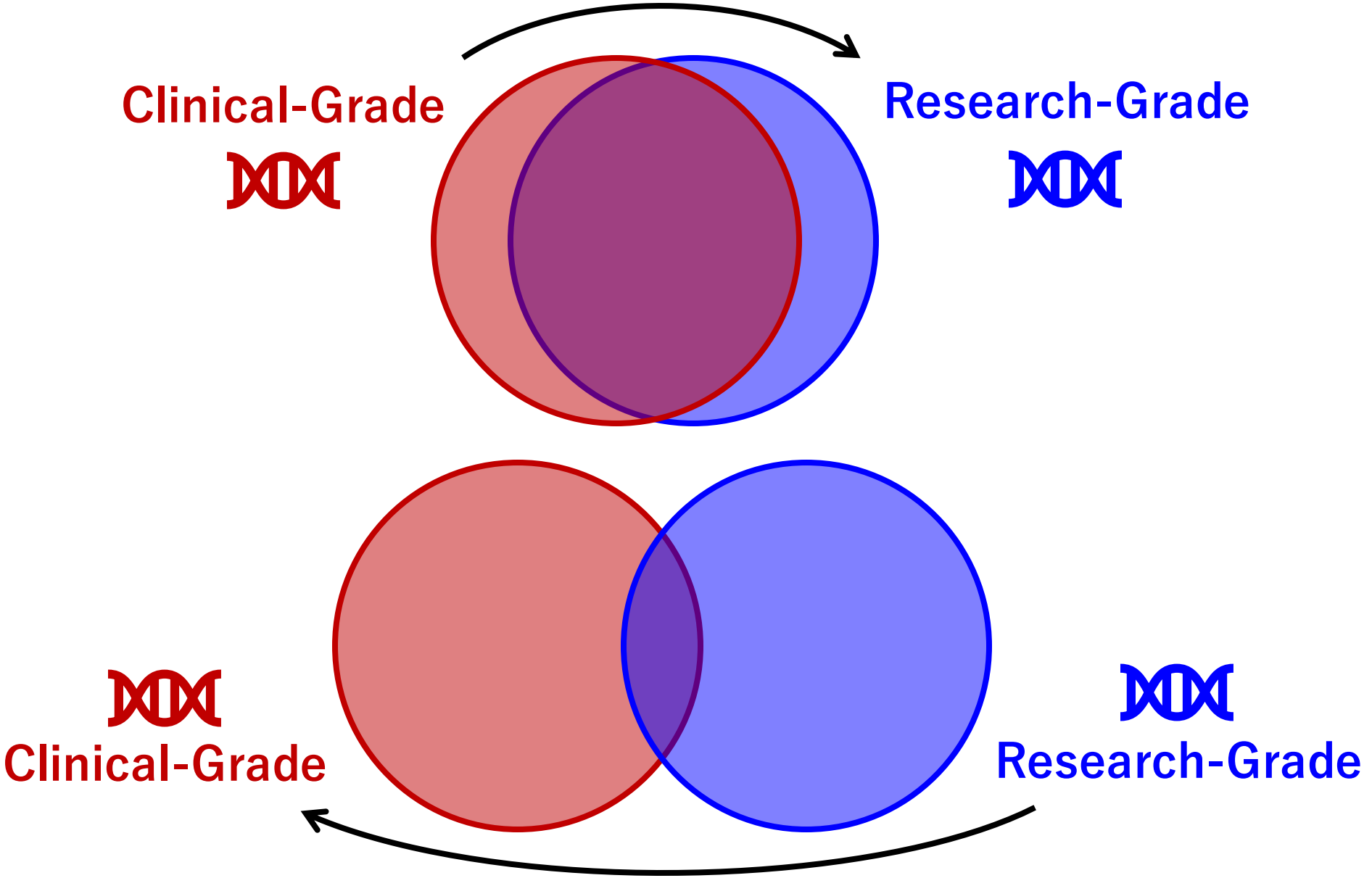
Genomic Data for Clinical Use (assumption)

- Format is use-case specific
- Whole Genome Sequence (WGS) data
- Large Projects: 1+ Million Genomes (EU) [1]



[1] https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes

Genomic Data for Clinical Use (reality)



© Elias Crum 2025

The current landscape

Intended use is VERY important

Clinical use ≠ Research use

** Funding goes predominantly to RESEARCH-GRADE **

Cheaper

Fewer regulatory challenges

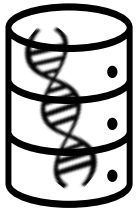
Funders / Policy makers misunderstanding

** But how do we “translate” to clinical use?? **

© Elias Crum 2025

Sharing Genomic Data – Technology Considerations

Key Implementation Factors



Data Storage

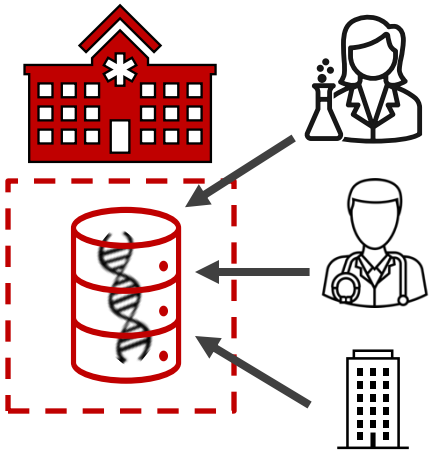
Considerations:

- patient consent
- data security
- access controls

&

Considerations:

- consent changes
- governance
- infrastructure

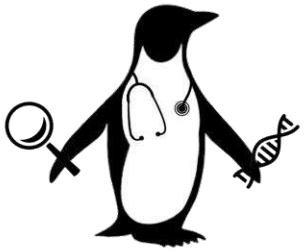


Data Sharing

A more in-depth analysis of these challenges + potential solutions detailed in an upcoming publication in *Computers in Biology and Medicine*

© Elias Crum 2025

PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice



Genomic Data → Clinical Care

Framework + App

Semantic Representation

Query Capabilities

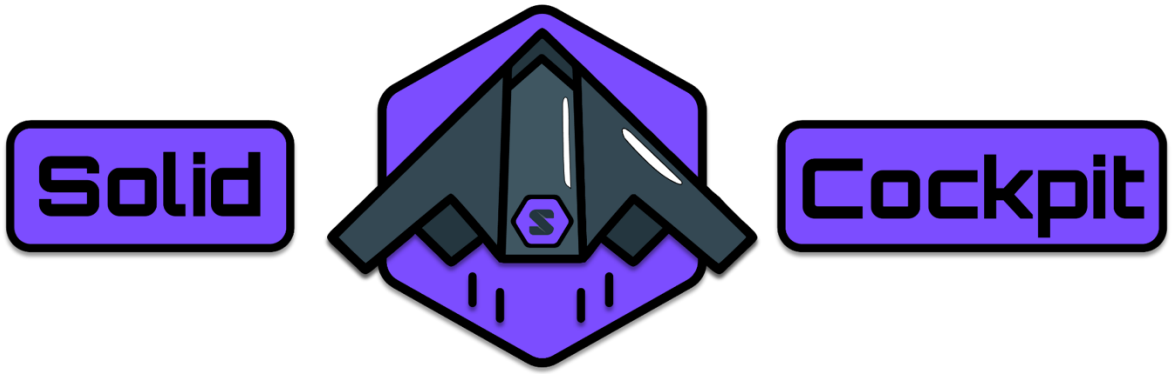
1. Genomic Data Interaction

2. Semantify Genomic Data

© Elias Crum 2025

1. Data Interaction Framework

A Web Application to:



- *Upload Data to a Solid Pod*
- *Browse contents of a Solid Pod*
- *Edit Privacy of (and share) data in a Solid Pod*
- *Query Data in Solid Pods + SPARQL Endpoints*
  - *Includes a query cache for storing queries and results*

*** Enables patient, provider, and auditor control over data ***

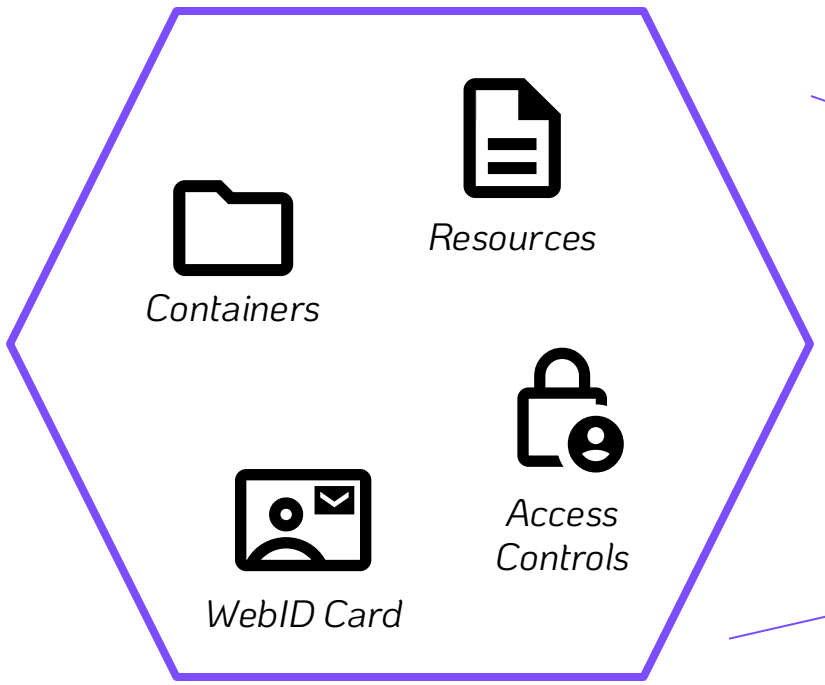
© Elias Crum 2025

What is a Solid Pod?

Secure, Decentralized Data Storage Specification

- *Pod exists on a Server (accessible to the Web)*
- *Can store files, RDF data, and Linked Data*
- *Provides control policies for stored data*

Elias' Pod





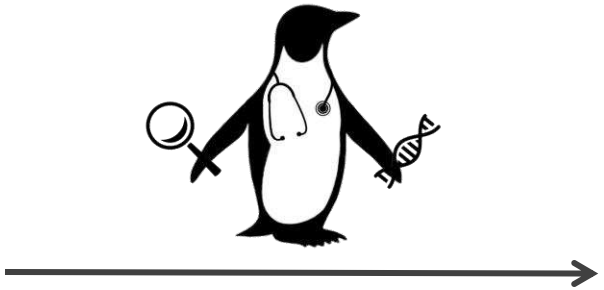
*** Enables secure, authenticated sharing via HTTP protocols ***

https://solidproject.org/about

PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice



Genomic Data





Clinical Care

Framework + App

Semantic Representation



Query Capabilities

1. Genomic Data Interaction

2. Semantify Genomic Data

© Elias Crum 2025

2. Genomic Data Representation

From a file-based approach

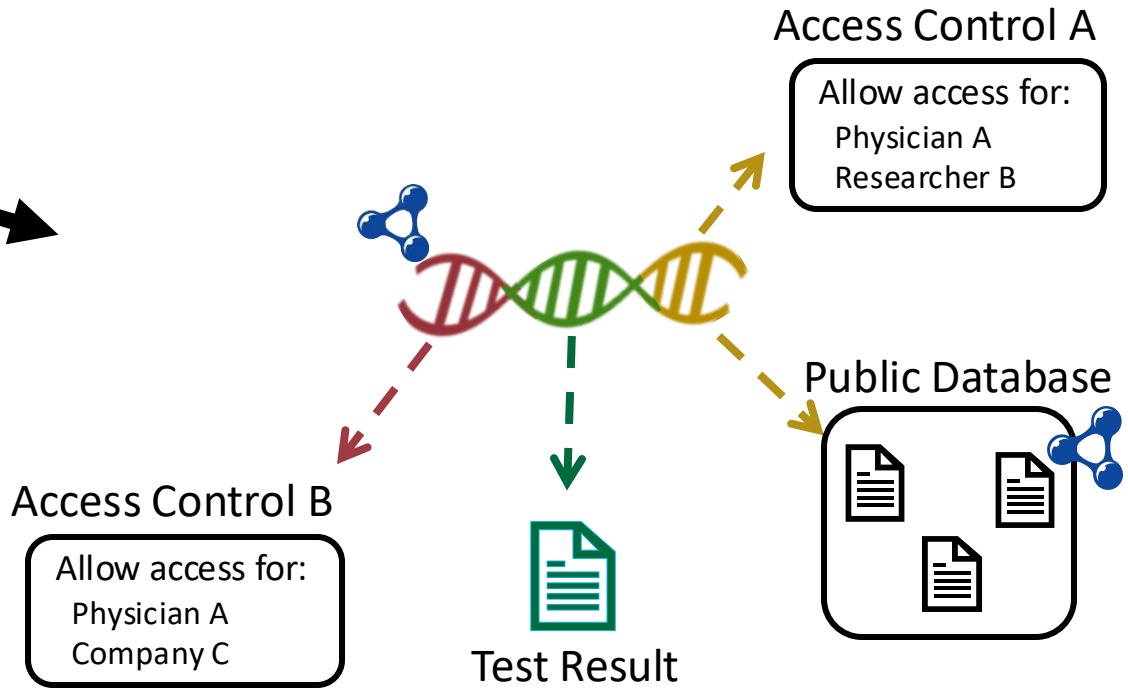


Conversion using existing ontologies

Policy definition via ODRL[1]

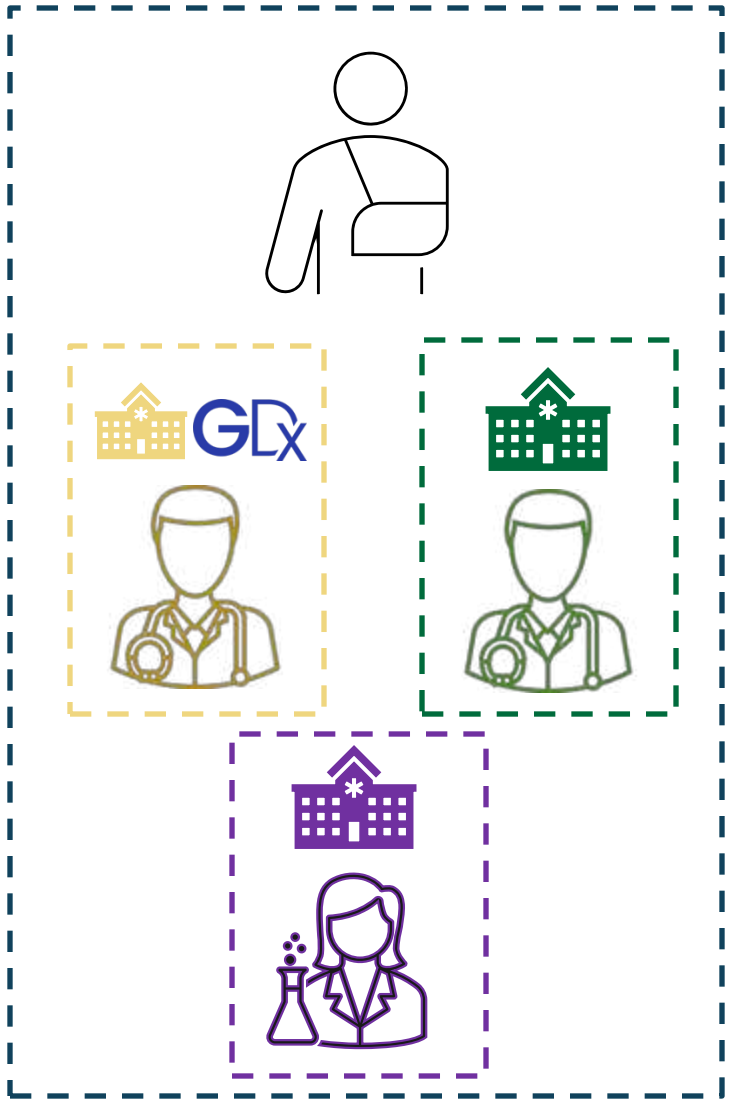
Implementation within Solid Pod Env

To a granular, dynamic RDF approach



[1] https://www.w3.org/TR/odrl-model/

Future Possibilities?

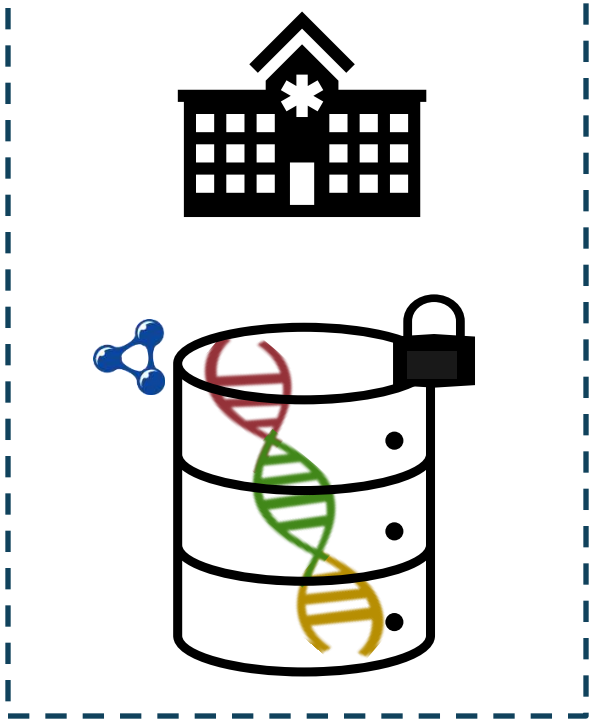


Data Users

Technological Infrastructure



Data Storage

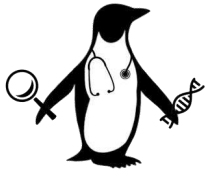


MORE dynamic

MORE patient in-the-loop

© Elias Crum 2025

Thank you!



More PhD specifics

LinkedIn

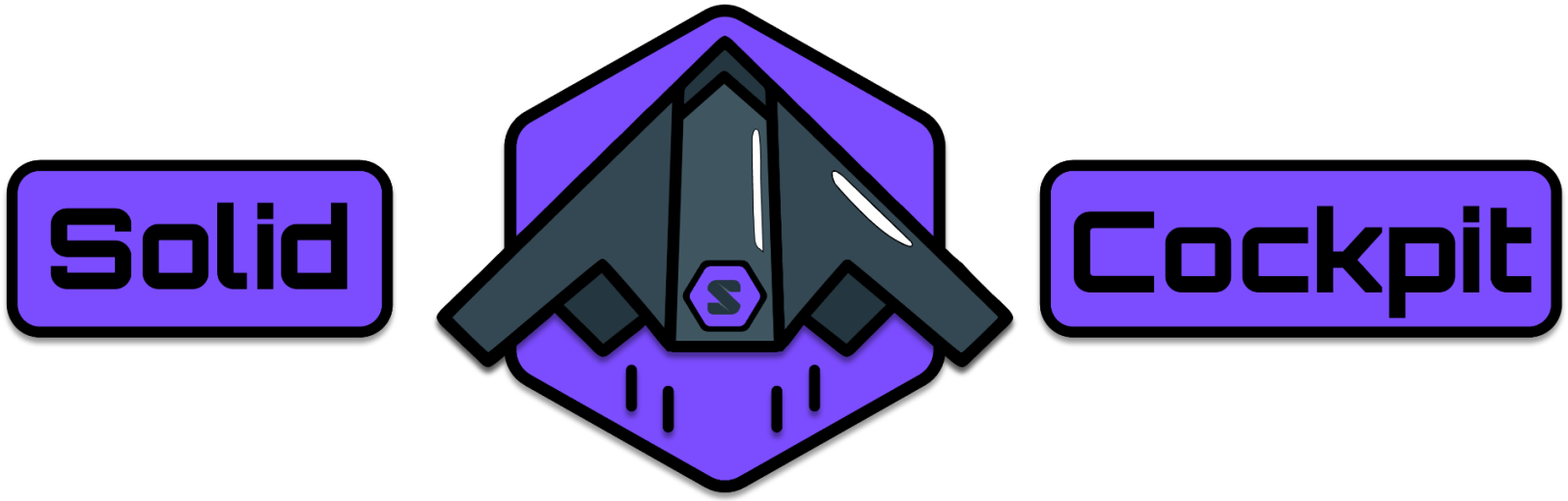
GitHub

Advisors: Dr. Gökhan Ertaylan[1], Dr. Bart Buelens[1], Dr. Ruben Taelman[2], Prof. Dr. Ruben Verborgh[2] 1 = vito 2 = GHENT UNIVERSITY

© Elias Crum 2025

Informal Demo / Tutorial Later Today

(for those interested)



knowledgeonwebscale.github.io/solid-cockpit/

© Elias Crum 2025

Supplemental Slides

© Elias Crum 2025

Genomic Data

Clinical-Grade

Higher read quality

More expensive

Stricter use-regulations

Research-Grade

Lower read quality

Less expensive

Flexible use-regulations

Dimension	Clinical Grade Genome Data Sharing	Research Grade Genome Data Sharing
1. Primary Purpose and Scope	Patient care and medical decision-making.	Discovery and hypothesis-driven studies.
2. Data Accuracy and Quality	Stringent quality control (QC) and validation.	High but no absolute QC standards.
3. Regulatory Requirements	Subject to medical regulatory frameworks.	Subject to research ethics regulations.
4. Consent Requirements	Clinical-informed consent.	Research-informed consent.
5. Privacy and Confidentiality	Highest level of patient privacy required.	Strong but variable privacy measures.
6. Security Controls	Robust, enterprise-grade security.	Secure but more flexible.
7. Data Governance and Oversight	Hospital/clinic-based governance.	Institution or consortium-based governance.
8. Ethical Considerations	Focus on direct patient benefit and “do no harm.”	Focus on scientific advancement and participant welfare.
9. Infrastructure Requirements	Highly reliable, clinically validated systems.	Flexible, scalable research infrastructure.
10. Architecture Requirements	Standardized, interoperable clinical systems.	Flexible, modular architectures.
11. Legal Liabilities	High.	Moderate.
12. Interoperability	High necessity for clinical integration.	Variable.
13. Data Ownership and Control	Patient-oriented ownership with institutional custodianship.	Institution or sponsor-owned.
14. Data Sharing Scope	Limited to necessary clinical stakeholders.	Broad or consortium-based.
15. Cost and Funding Models	Healthcare reimbursement or institutional budgets.	Grant-funded or institutional research budgets.

Genomic Sequence Data Sharing for Clinical Practice: A Scoping Review

vito

GHENT UNIVERSITY

Aim: Identify existing implementations, guidelines, and challenges of clinical sharing of genomic data.

Methods: Reviewed 55 studies, 57 national genomics projects, and 20 major genomics companies.

Results: Clinical genome sharing implementations exist, yet scalability and widespread adoption remain limited.

Suggested steps forward:

Harmonize legal bases for secondary clinical use of research data



Mandate (GA4GH) technical standards



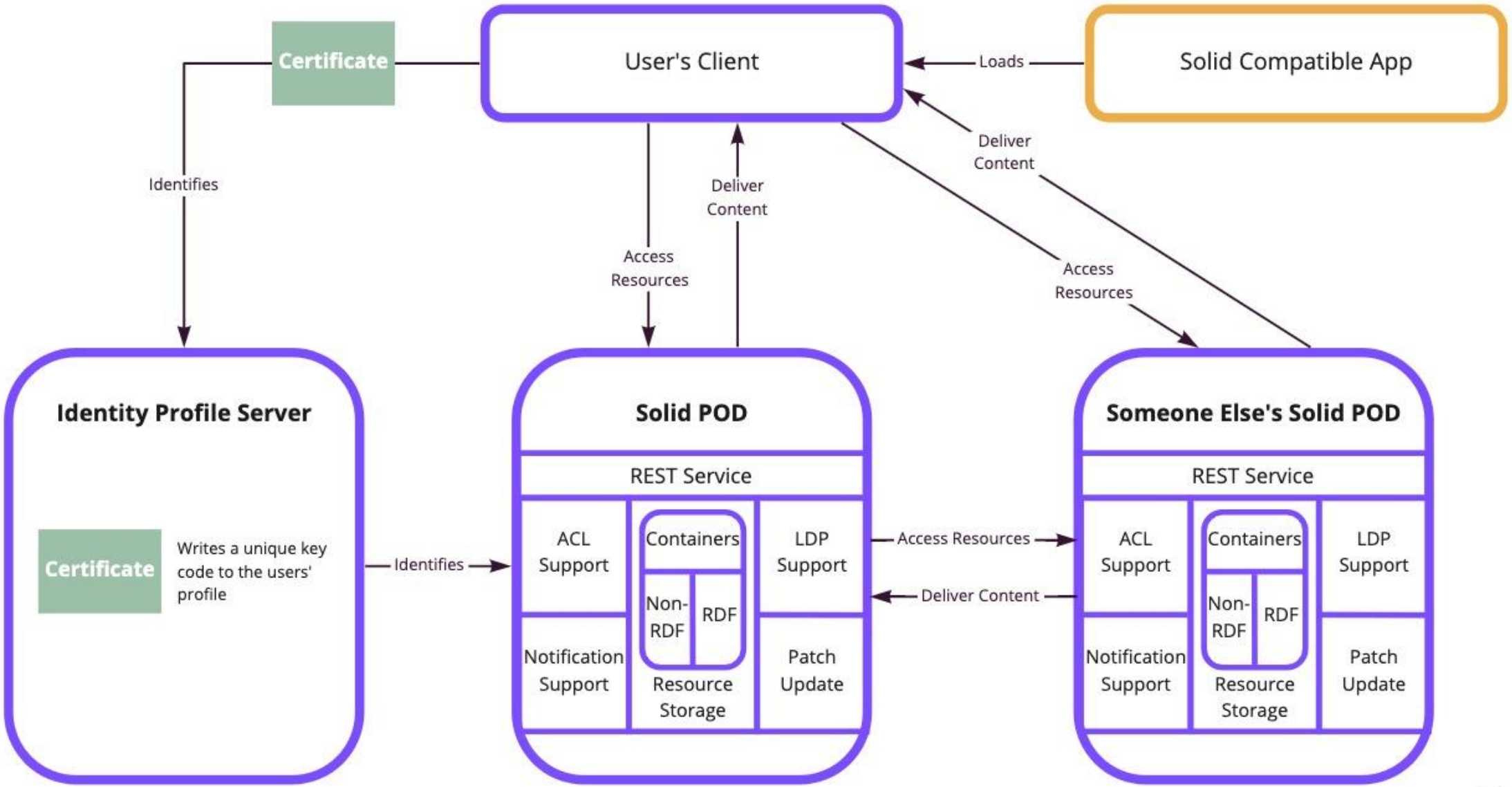
Scale laboratory accreditation & continual pipeline validation



Align economic incentives



Solid's Architecture



miro

© El https://solid.georgetown.domains/technical-component/

Solid Pod Implementations?

BBC MyPDS App

athumi

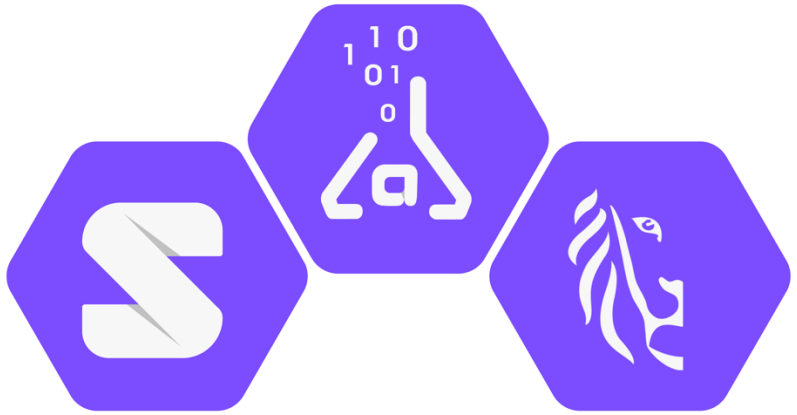
Flemish Data Utility Company

we are

health data vaults

Flanders
State of the Art

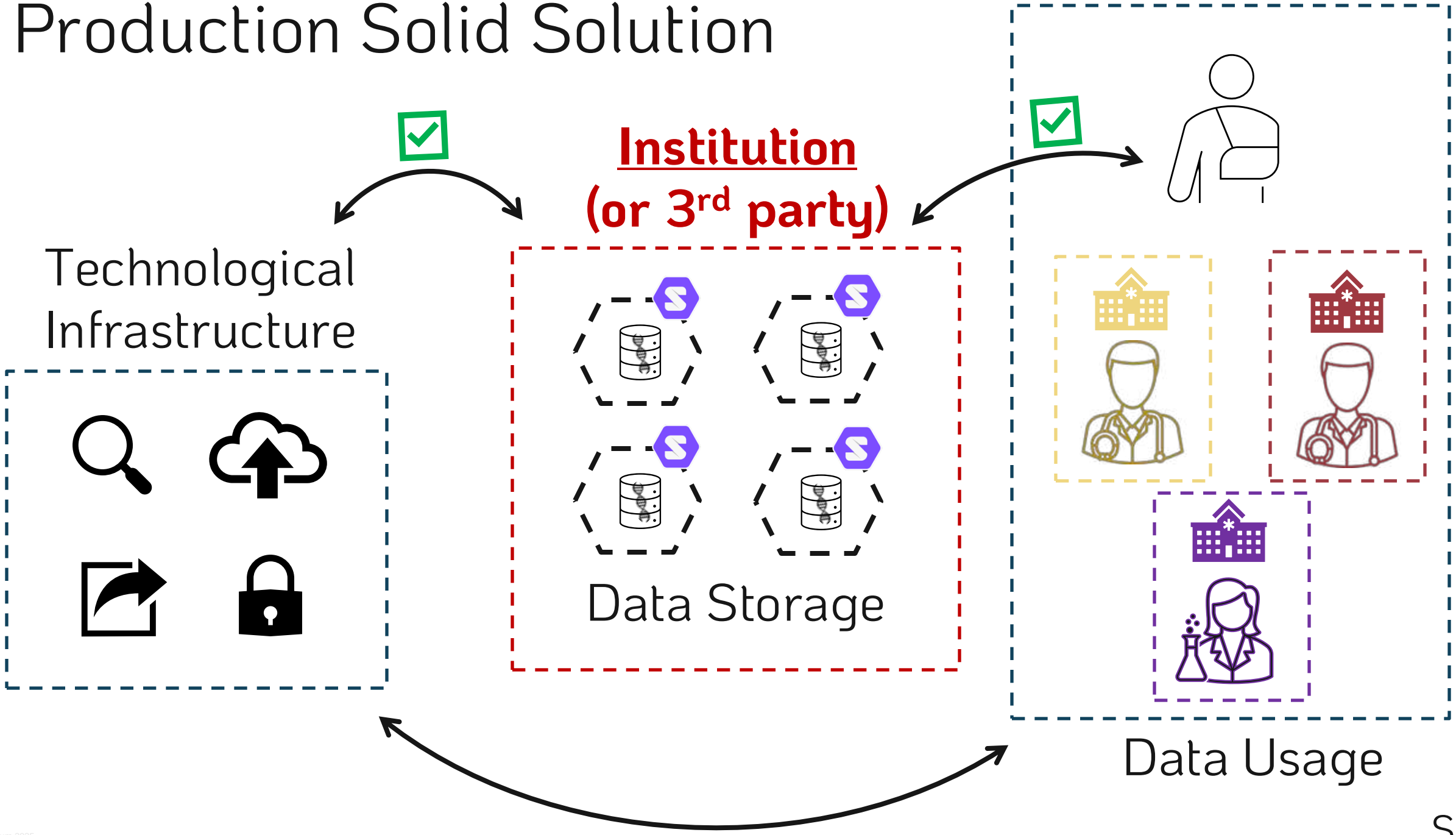
Flemish Smart Dataspace



Solid Lab Use-cases

© Elias Crum 2025

A Production Solid Solution



© Elias Crum 2025

Semantic VCF *– Sequencing Data Formats Intro*

FASTQ ✚ **FASTA** → **SAM/BAM** → **VCF**

Raw Sequencing Outputs

Contiguous DNA sequence

FASTQ reads aligned to a FASTA reference

Areas in SAM where sample and reference differ

```
@FORJUSP02
CCGTCAATTC
+
AAAAAAAAAA
```

```
1  >gi|568336023|gb|CM000663.2| Homo sapiens chromosome 1, GRCh
2  CCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACC
3  ACCCTAACCCTAACCCTAACCCTAACCCAACCCTAACCCTAACCCTAACCCTAACCCTAA
4  TAACCCTAACCCTAACCCTAACCCTAACCTAACCCTAACCCTAACCCTAACCCTA
5  CTAACCCTAACCCCTAACCCTAACCCTAACCCTAACCCTA
6  CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCTAACCC
7  CCTAACCCTAACCCTAACCCTAACCCTAACCCCTAACCCCTA
8  TAACCCTAACCCCTAACCCTAACCCTAACCCTAACCCTCGCG
9  TCTGACCTGAGGAGAACTGTGCTCCGCCTTCAGAGTACCACC
10 CGCCCTCGCGGTGCTCTCCGGGTCTGTGCTGAGGAGAACGCA
11 CGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGCA
12 AGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGCC
13 CGCGCCGGCGCAGGCGCAGACACATGCTAGCGCGTCGGGGTG
14 GCCGCGCCGGCGCAGGCGCAGAGACACATGCTACCGCGTCCA
15 AGGCGCACCGCGCCGGCGCAGGCGCAGAGACACATGCTAGCG
16 GCAGAGACGCAAGCCTACGGGCGGGGGTTGGGGGGGCGTGTG
17 GCTGGGGCGGGGGGAGGGTGGCGCCGTGCACGCGCAGAAACT
18 GGTAGAACCTCAGTAATCCGAAAAGCCGGGATCGACCGCCCC
19 GCTTGCTCACGGTGCTGTGCCAGGGCGCCCCCTGCTGGCGAC
20 GAGTGGTGGCCAGCGCCCCCTGCTGGCGCCGGGGCACTGCAG
```

Q scores (as ASCII ch

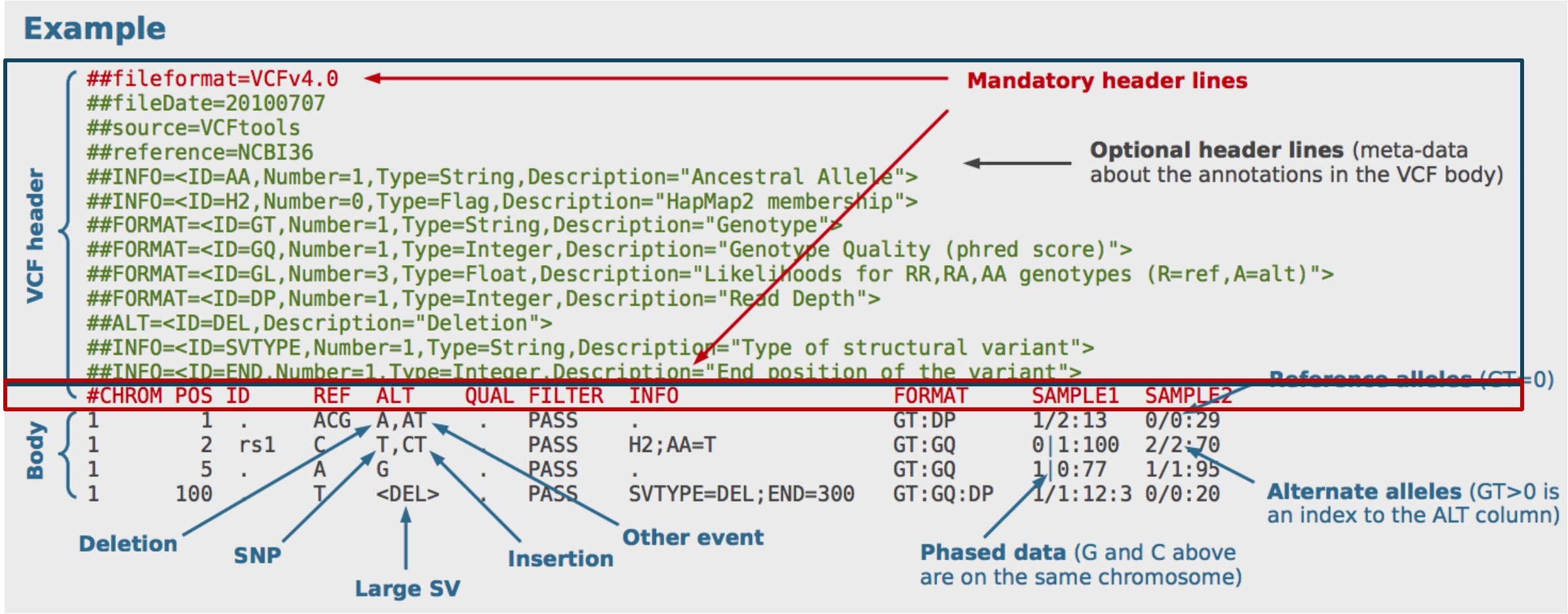
Base=T, Q=':'=25

```
@HD     VN:1.0  SO
@SQ     SN:chr20
@PG     ID:TopHat
lign-edit-dist 2 -
20 /data/user446/m
HWI-ST1145:74:C101
       CCGTGTTTAAA
C    BBDCCDDCCDDDD
    AS:i:-15
HWI-ST1145:74:C101
       TGCTGGATCAT
G    DCDDDDEDDDDDD
    AS:i:-16
HWI-ST1145:74:C101DACXX:7:1204:14760:4030       16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699       0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
accepted_hits.sam
```

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID        REF    ALT     QUAL FILTER INFO                              FORMAT      NA00001        NA00002        NA00003
20     14370   rs6054257 G      A       29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20     17330   .         T      A       3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3   0/0:41:3
20     1110696 rs6040355 A      G,T     67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2  2/2:35:4
20     1230237 .         T      .       47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20     1234567 microsat1 GTC    G,GTCT  50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4       0/2:17:2       1/1:40:3
```

© Elias Crum 2025

What does a VCF File Look like?



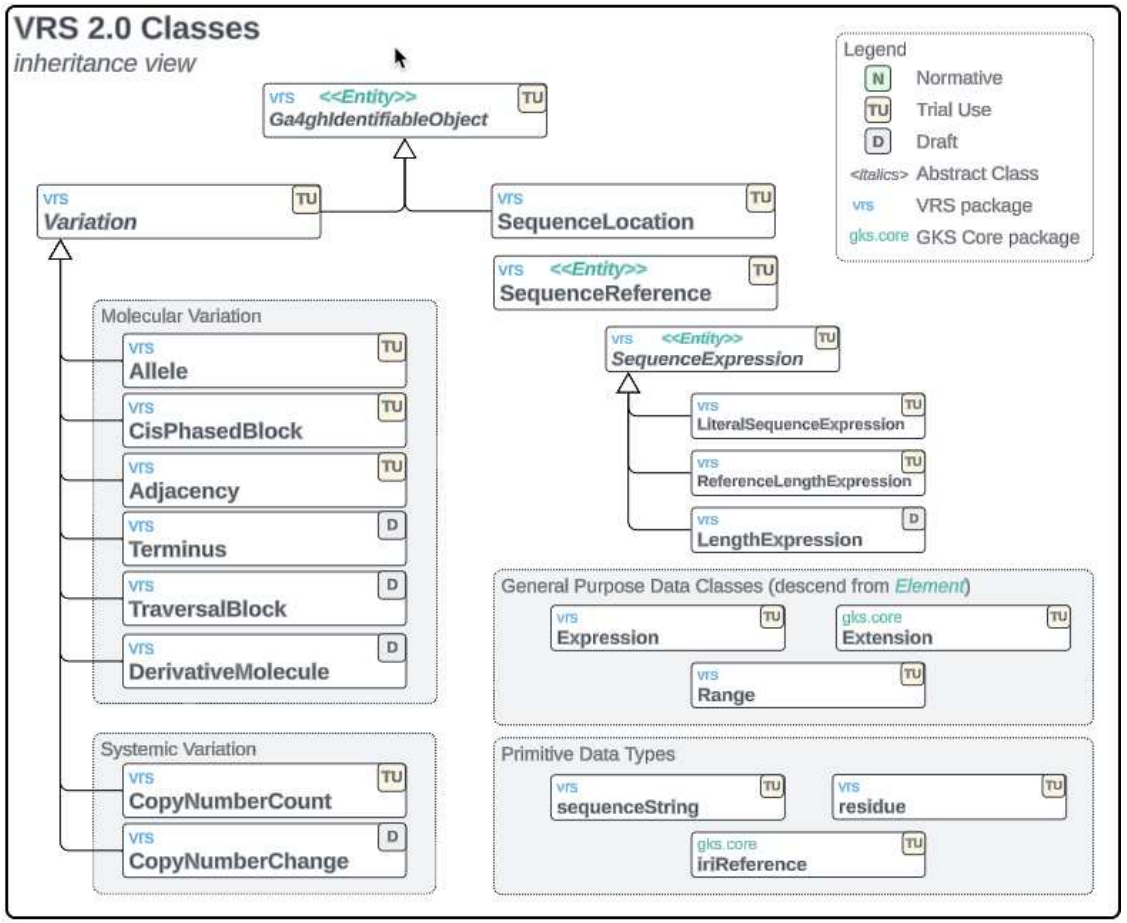
© Elliot Drum 2025

https://davetang.github.io/learning_vcf_file/

VCF Spec: https://samtools.github.io/hts-specs/VCFv4.2.pdf

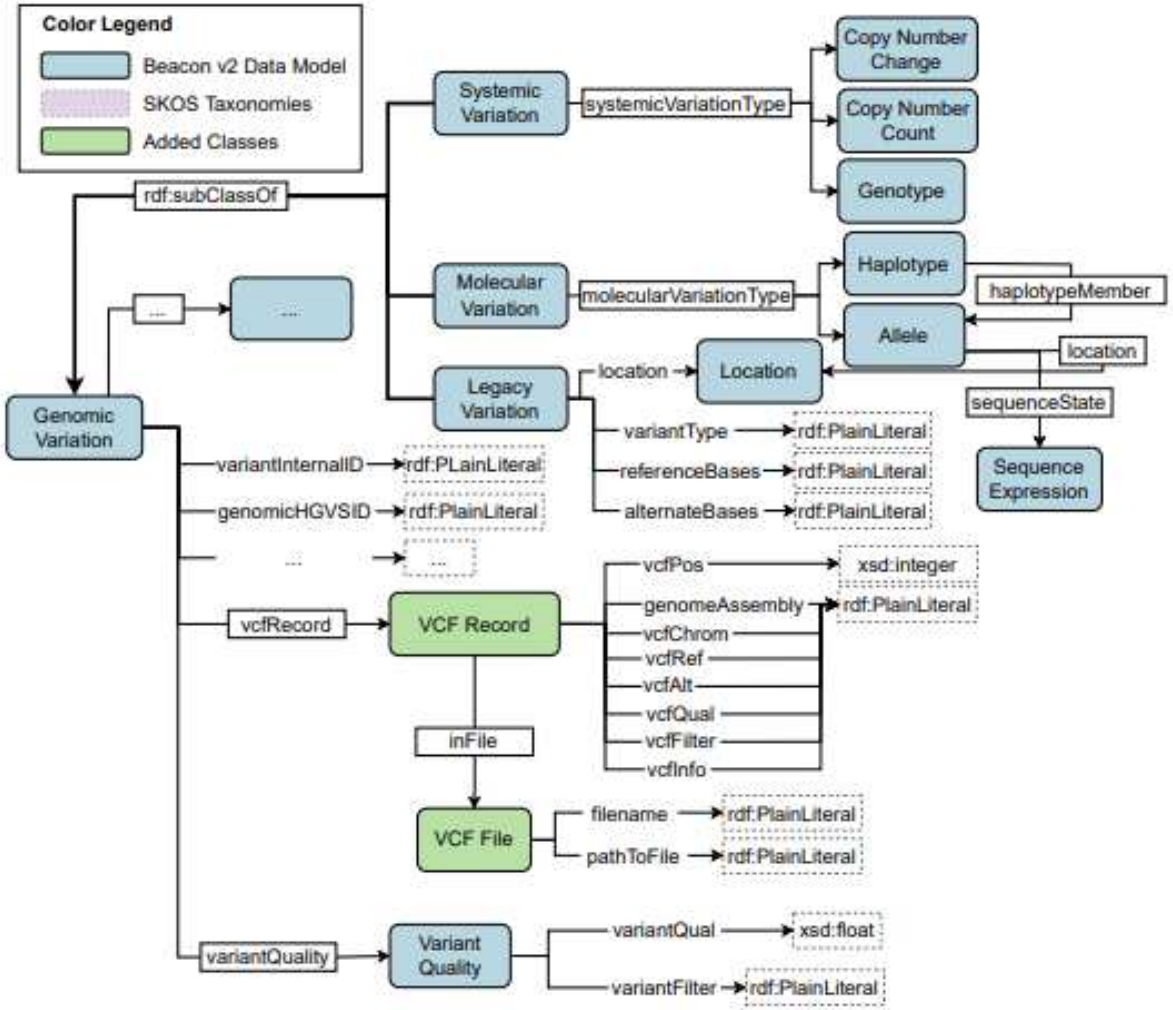
Representing VCF Semantically?

GA4GH Variant Schema:



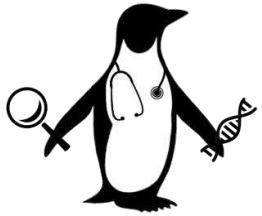
https://vrs.ga4gh.org/en/2.0.0-ballot.2024-11/appendices/class_diagram.html#classdiagram

HERO Genomics Ontology:



https://www.dei.unipd.it/~silvello/papers/2025_MCS_SWAT4HCLS.pdf

Envisioned Product Evolution











PhD End

Short Term

Medium Term

Long Term

PhD Deliverable

Start-up / Product development + Hospital implementation

Integrate with EHR + Increase application domains

Increase 3rd party application support + Scale to Europe and beyond

© Elias Crum 2025